NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA
Department of Computer Science & Engineering
End-Semester Examination, Spring 2024-25
B. Tech. 6th Semester

Subject Name: Computer Networks **Maximum Marks: 50**
Subject Code: CS3002 **Exam Duration: 3 Hours**
Number of pages: 2

Note: There are ***Six*** questions. Attempt all the questions. All parts of the same question must be attempted in a sequence.

Q.1 (a) Consider the queuing delay in a router buffer, where the packet experiences a delay as it waits to be transmitted onto the link. Assume a constant transmission rate of R = 1200000 bps, a constant packet-length L = 5500 bits, and the average rate of packet arrival a = 87 packets/second. Calculate the traffic intensity, queuing delay, and average delay in ms.?

(b) Which Layer in OSI Model is responsible for Error control? List at least three error detection techniques. What is the hamming distance between the words 1010101, 1100110?

(c) Define the Cookies and Web Cache with their application.

[3+4+3=10Marks, (a):CO1(L3); (b)&(c):CO1(L2)]

Q.2 Consider the following plot (Fig. 1) of TCP window size as function of transmission round. Answer the following questions, in all cases you should provide a short discussion to justify the answer.

- **(a)** What is the value of threshold at 18th transmission round?
- **(b)** Identify the interval of time when TCP slow start is operating
- **(c)** Assuming a packet loss is detected after 26th round by receipt of a 3 duplicate ACK, what will be the values of the congestion window size and threshold?
- **(d)** Find average throughput from 6 to 16 transmission rounds when RTT=25.6 ms and MSS=512 Bytes.

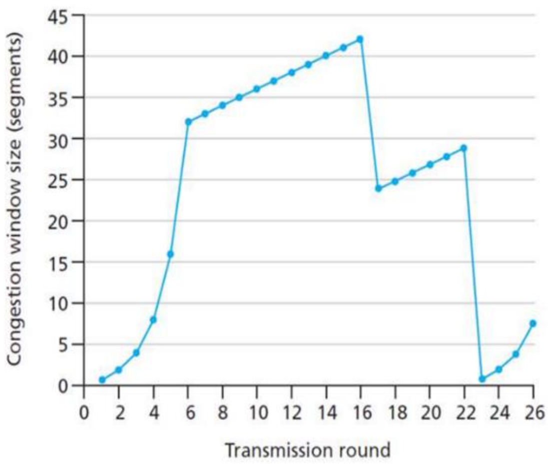

Fig. 1

[2+2+3+3=10 Marks, CO3(L4)]

Q.3 An ISP company X has a class A network 28.0.0.0 and the company provides 9024 subnets in its network. Each subnet has its own requirements, but the largest subnet needs 580 host addresses. Design the IP addressing and answer the following:

(a) What is the subnet mask and the maximum number of hosts per subnet?

(b) How many bits for the network ID, Subnet ID, and host ID?

(c) What is the layout of the addresses provided by company X? Specify the first and last IP addresses assigned to the host machines for the first four subnets.

[2+2+4=8 Marks, CO4(L5)]

Q.4 An IP datagram carrying 6000 bytes of data each must be sent over a links as follows: S--A—B—C, where S is a source and C is a destination. Link AB has an MTU of 3020 bytes and link BC has an MTU of 2020 bytes. Assume the datagram has IHL = 5; Identification number 201. How many fragments will be generated? Examine and determine the IP datagram fields after the fragmentation. Construct a table and answer values of the following fields for each of the fragments for AB and BC links: Identification, Total Length, D-bit, M-bit, and Fragmentation Offset.

[6 Marks, CO(L5)]

Q.5 Consider the network scenario and costs to each link as shown in Fig. 2. Assume that each node knows their routing table initially [Use Distance Vector Routing].

(a) Now the link between nodes A and D is broken then update the routing table at node A. Show all steps involved in computing the routing table at A.

(b) If node D receives the distance vectors from node B, C, E, and G are (7, 0, 3, 6, 2, 7, 10), (6, 4, 0, 7, 3, 5, 9), (3, 7, 5, 4, 0, 6, 11), and (2, 4, 5, 3, 6, 7, 0), respectively. Node D knows the costs to its neighbors as per the distances depicted in Fig. 2 where link A-D is already broken. The distance vectors are distances for nodes A, B, C, D, E, F, G. Determine the routing table at node D by showing the intermediate steps.

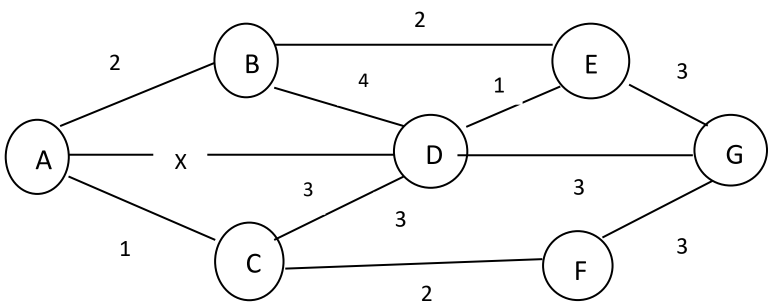

Fig. 2 Network Topology

[3+3=6 Marks, CO4(L3)]

Q.6 (a) A slotted ALOHA network transmits 200-bit frames using a shared channel with a 200 Kbps bandwidth. Find the throughput of the system, if the system (all stations put together) produces 250 frames per second.

(b) CSMA/CD technique is applied to wired or wireless medium communication? Justify your answer. Draw the flowchart for CSMA/CD technique.

(c) Consider the Cyclic Redundancy Check (CRC) algorithm. Suppose that the 4-bit generator (G) is 1001, that the data payload (D) is 10011010 and that r = 3. What would be the CRC bits (R) associated with the data payload D?

[3+3+4=10 Marks, CO5(L3)]